FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

CONCEPT RECAPITULATION TEST – III

PAPER –2

TEST DATE: 26-04-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A ***(01 – 04, 19 – 22, 37 – 40)*****:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A ***(05 –10, 23 – 28, 41 – 46)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B ***(11 – 18, 29 – 36, 47 – 54)*****:** This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

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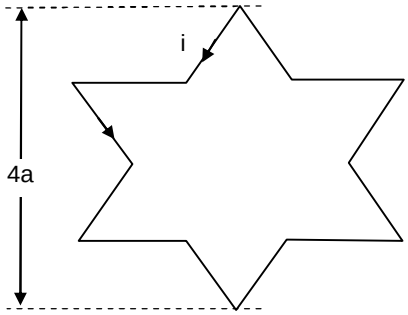
Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. What is magnetic dipole moment of the loop?



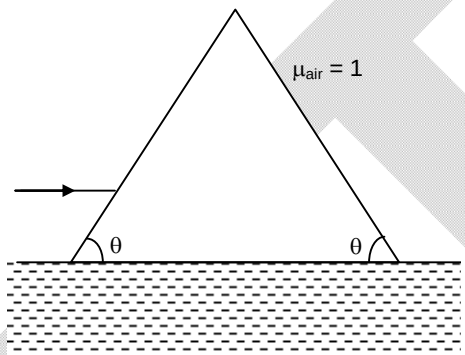
(A) $4a^2i$

(B) $4\sqrt{3}\,a^2\,i$

(C) $8\sqrt{3}\,a^2\,i$

(D) None of these

2. An isosceles prism of base angle θ is kept on water surface $(\mu_\omega = 4/3)$. A ray of light parallel to base strikes prims as shown in figure. What should be minimum value of tan θ so that light ray emerges from other face after suffering total internal refection at base. (refractive index of prism is 7/4)



(A) $\sqrt{\frac{17}{21}}$

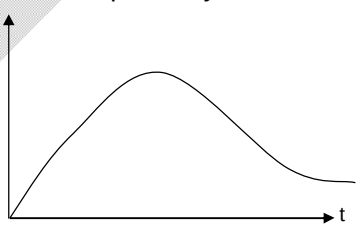
(B) $\frac{4}{\sqrt{185}}$

(C) $\frac{2}{\sqrt{201}}$

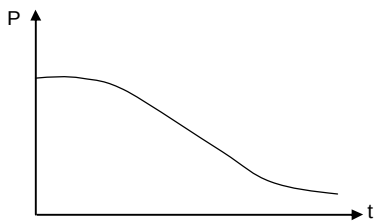
(D) $\sqrt{\frac{7}{201}}$

3. A particle of mass m (at rest) is subjected to a force $F = F_0e^{-bt}$ along the x-axis. Power delivered by the force is depicted by which of the following groups.

(A)

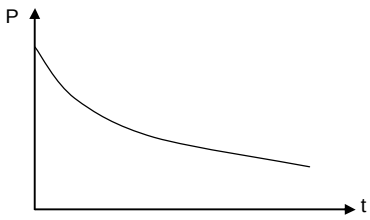


(B)

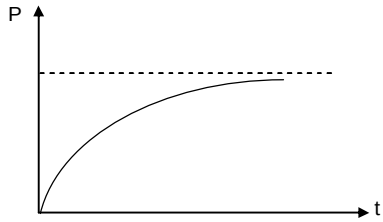


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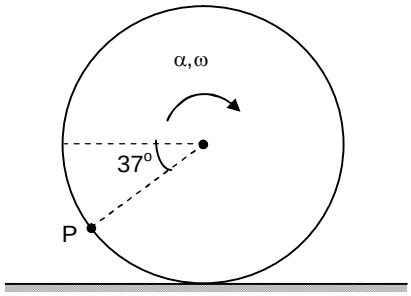
(C)



(D)



4. A ring of radius 20 cm is rolling on a rough horizontal surface. At a particular moment its angular speed is 5 rad/sec and angular acceleration is 25 rad/sec^2. Find acceleration of point P on it.



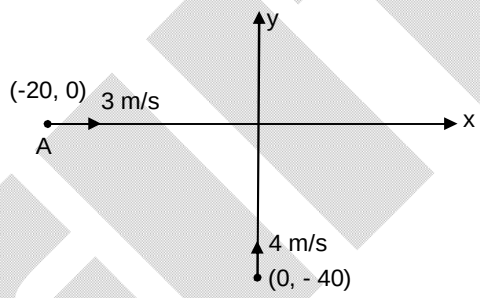
(A) $6\hat{i}+4\hat{j}$ (B) $6\hat{i}+7\hat{j}$

(C) $12\hat{i}+\hat{j}$ (D) $12\hat{i}+7\hat{j}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. Two particles A and B are moving along x-axis and y-axis respectively with speeds 3 m/s and 4 m/s as shown. Their initial distance (at t = 0) are 20 m & 40 m respectively from origin. Then choose the correct option(s)



(A) shortest distance between then is 18 m (B) shortest distance between then is 8 m

(C) Time to reach shortest distance is 8.8 sec (D) Time to reach shortest distance is 12 sec

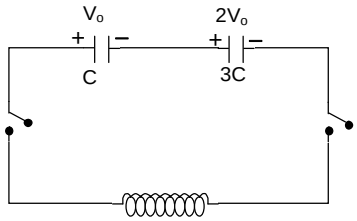
6. Which of the following forces are conservative is nature?

(A) $x^2\hat{i}+xy\hat{j}$ (B) $\left(x^2-y\right)\hat{i}+\left(y^2-x\right)\hat{j}$

(C) $x^2\hat{i}-y^2\hat{j}$ (D) none of these

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7. A given circuit is connected at t = 0 then, choose the correct option(s):



(A) maximum charge on capacitor C is $\frac{13CV_0}{4}$

(B) minimum charge on capacitor 3C is $\frac{7CV_0}{4}$

(C) maximum current through inductor is $\frac{3\sqrt{3}V_0}{2}\sqrt{\frac{C}{L}}$

(D) All of these

8. Current density in a cylindrical wire is varying with radial distance r as $J=\frac{J_0}{R^2}r(R-r)$. Where R is radius of wire. If B is magnetic field due to this wire. Then choose the correct option(s).

(A) maximum magnetic field will be at a distance $\frac{8R}{9}$

(B) maximum magnetic field will be at a distance R

(C) maximum magnetic field will be $\frac{\mu_0 J_0 R}{12}$

(D) maximum magnetic field will be $\frac{64\mu_0 J_0 R}{729}$

9. Electric potential in a region varies only with x-coordinate and is given by $v=2x^3$. Then choose the correct option(s).

(A) electric field in the region is $6x^2\,N/C$ along positive x-axis.

(B) electric field in the region is $6x^2\,N/C$ along negative x-axis.

(C) charge density in the region is constant with x-coordinate

(D) charge density varies linearly with x-coordinate

10. A particle of mass π kg moves along x axis in a conservative force field. Its potential energy is given by $U(x)=2x^3-9x^2+12x$ where all quantities are in SI units. If the particle is displaced from its stable equilibrium position it oscillates. Then choose the correct option(s).

(A) Maximum kinetic energy the particle can possess is 1 J.

(B) Maximum kinetic energy the particle can possess is 5 J.

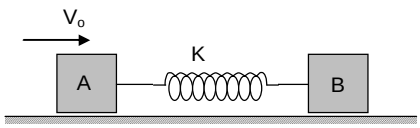
(C) Time period of small oscillations would be 1 sec.

(D) Mean position of the particle is x = 1 m.

Section – B (Maximum Marks: 24)

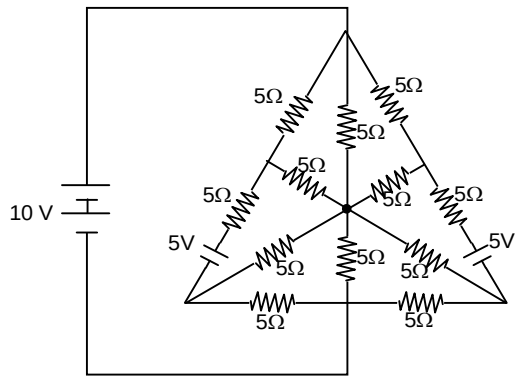
*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. Block A of mass 2 kg is given velocity $v_0=10\,m/s$ towards another block of mass 4 kg. If spring constant of ideal spring is 150 N/m. Find minimum speed attained by block A.



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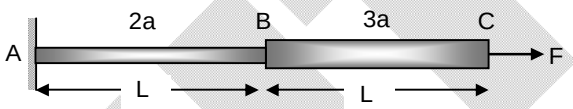
12. Find the current provided by 5V cell in the adjacent circuit (internal resistance of each cell is zero).



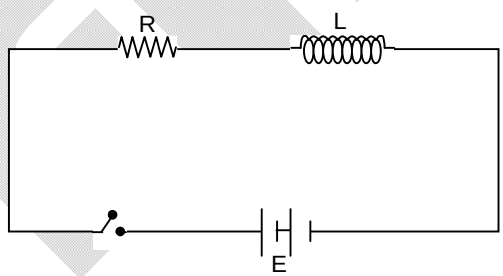
13. A particle of mass m is kept in potential field $U = \frac{U_0}{2}\left[1-\frac{x^2}{a^2}\right]^2$. Particle is at its stable equilibrium position. If it is displaced slightly it executes SHM with time period $T = \pi\sqrt{\frac{a^2 m}{xU_0}}$. Find x = ?

14. The coefficient of thermal expansion of a rod is temperature dependent and is given by formula $\alpha = \frac{T}{T_0}$, where T_0 is constant and T is temperature. If the length of rod is L at temperature 0^oC, then temperature at which its length doubles is given by $\sqrt{T_0 \ln(x)}$. Find x = ?

15. Two steel wire of same length but radii 2a and 3a are connected together end to end and tied to wall as shown. The force stretches the combination by 13 mm. Find the displacement of point B. (Ans. in mm)



16. A resistor and an inductor are connected at t = 0 through a battery. Maximum current through the circuit is I_o. When power across inductor is maximum current through the circuit is I_1. Find $\frac{I_0}{I_1}$.



17. The activity of a sample reduces from A_0 to $\frac{A_0}{\sqrt{2}}$ in 10 minutes. The activity of the sample after 30 more minutes will be A_1. Find $\frac{A_0}{A_1} = ?$

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18. A screen is placed at very large distance from two coherent sources S_1 and S_2 separated by distance d. There are two points A & B on screen. $\angle AOB = 0.12°$. Find number of points of total darkness on line segment AB. (Given $\lambda / d = 10^{-3}$)



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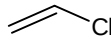
Chemistry

PART – II

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

19. Pick the correct statement, from the following:

(I)  : PVC

(II) $HOOC-(CH_2)_4-COOH, H_2N-(CH_2)_6-NH_2$: Nylon – 6, 6

(III) C_6H_5OH, HCHO : Bakelite

(A) Only I is correct
(B) II and III are incorrect
(C) Only III is incorrect
(D) I, II and III all are correct

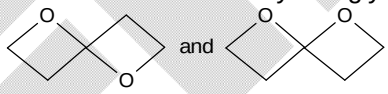
20. 1.45g of acetone dissolved in 100 g of benzene gave a solution which freezes at 277.1 K. Pure benzene freezes at 278.4 K. 7.81 g of a solid (X) dissolved in 100 g of benzene gave a solution which froze at 275.8 K. The molecular weight of X is
(A) 156.0
(B) 154.8
(C) 155.4
(D) 156.20

21. Gold number of a protective colloid is 0.04. Hence 200 mL of gold sol will require how many mg protective colloid so that gold sol is not coagulated by 1 mL of 10% NaCl solution?
(A) 0.40 mg
(B) 0.04 mg
(C) 0.80 mg
(D) 8 mg

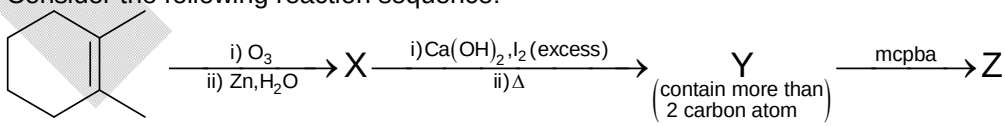
22. In which of the following, molecule is linear and central atom does not involve d orbital in hybridization?
(A) I_3^-
(B) SO_2
(C) O_3
(D) $\overset{\oplus}{N}O_2$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Which of the following statement is not correct?
(A) Butanoic acid and 2-methylpropanoic acid are position isomers.
(B) Most stable from of ethylene glycol is anti.
(C)  are geometrical isomers
(D) Only two isomeric monochloro derivatives are possible for propane

24. Consider the following reaction sequence:



Pick the incorrect statement(s):
(A) X contain two –COOH group
(B) X contain two $-COCH_3$ group
(C) Y contain six atom of iodine per molecule
(D) Z is epoxide

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25. Which of the following statements is/are incorrect?

(A) $t_{\frac{3}{4}} = 2 \times t_{\frac{1}{2}}$ (first order)

(B) $k = \text{time}^{-1}$ (first order)

(C) $t_{\frac{1}{2}} \alpha$ conc (zero order)

(D) $k = \text{time}^{-1}\text{mol}^{-1}\text{L}$ (zero order)

26. A solution has 0.001 M Mg^{2+} and 0.01 M NH_3. Further some NH_4Cl was added in 500 mL of above solution. K_{sp} of $Mg(OH)_2$ is 10^{-13}; base ionization constant of NH_3 is 2×10^{-5}.
Now pick the correct statement(s):
(A) The maximum pH at which there will be no precipitation of $Mg(OH)_2$ occur is 9
(B) Amount of NH_4Cl required to prevent the formation of $Mg(OH)_2$ in above solution is 0.535g
(C) 99% Mg^{2+} was precipitated at pH 10
(D) All above statements are wrong

27. Pick out the incorrect statement(s):

(A) NCl_5 does not exist

(B) $Ca(HCO_3)_2$ exist in solid state

(C) Total number of lone pair in I_2Cl_6 is 20

(D) Pb^{+4} is better oxidizing agent then Pb^{2+}

28. 50 gram of x% oleum is mixed with 10 gram of water. Resulting solution required 1.204 moles of NaOH for complete neutralization. What will be composition of mixture before neutralization?
(A) 60 gram H_2SO_4 only
(B) 59 gram H_2SO_4 and 1 gram water
(C) 54.5 gram H_2SO_4 and 5.5 gram water
(D) x = 118

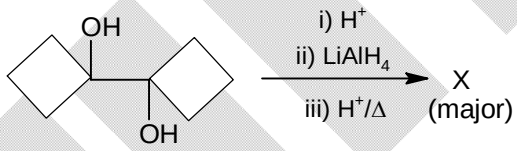
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

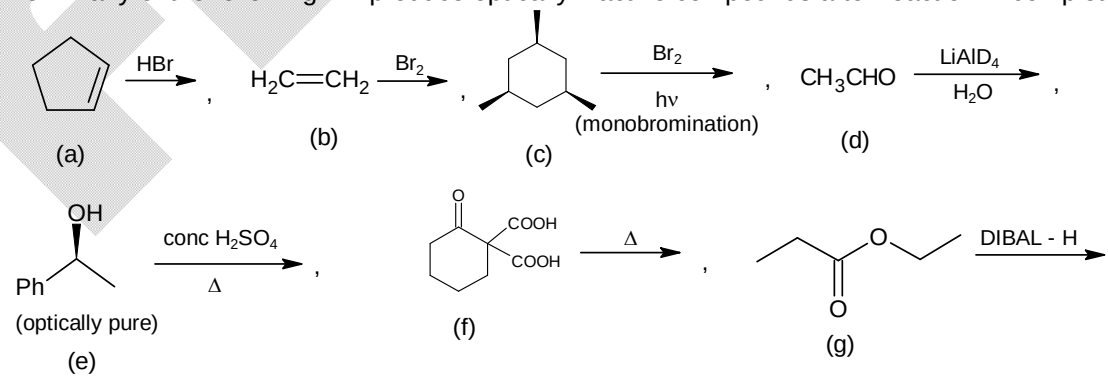
29. How many monobromo compounds (excluding stereoisomer) will give 2, 2, 3-trimethylpentane on treatment with magnesium in THF followed by water?

30. Find the number of moles of formic acid formed from one mole of fructose on treating with five moles of periodic acid.

31. The number of plane of symmetry in X is



32. How many of the following will produce optically inactive compounds after reaction in completed?



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33. The number of $N-M-N$ linkages in $\left[M(en)_3\right]^{n+}$ is 5x. Find x.
(M = Transition metal, en = ethylene diamine)

34. The ratio of rates of diffusion of CH_4 and SO_2, under identical condition is

35. A sample of three hydrogen atom all in fourth excited state, de excites to ground state. Calculate maximum number of spectral lines in spectra of above sample, if no line is seen Paschen series.

36. Find the value of (a + b + d) in given below balanced reaction:
$aMnO_4^- + bFe^{2+} + cH^+ \longrightarrow dMn^{2+} + eFe^{3+} + fH_2O$

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Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. Let f(x) and g(x) be continuous, positive function such that $f(-x)=g(x)-1$, $f(x)=\frac{g(x)}{g(-x)}$ and $\int_{-22}^{22} f(x)dx = 2022$, then the value of $\int_{-22}^{22}\frac{f(x)}{g(x)}dx$ is:

(A) 1011 (B) 2022
(C) 2023 (D) 1010

38. If the normal at a point P to the hyperbola meets the transverse axis at G, and the value of $\frac{SG}{SP}$ is 4, then the eccentricity of the hyperbola is (where S is the focus of the hyperbola)

(A) 2 (B) 3
(C) 4 (D) 6

39. Let A and B are square matrices of same order satisfying AB = A and BA = B, then $\left(A^{2022}+B^{2022}\right)^{2023}$ is equal to:

(A) (A + B) (B) 2022 (A + B)
(C) $2^{2022}(A+B)$ (D) $2^{2023}(A+B)$

40. Given that $m,n,s,t\in(0,\infty), m+n=3, \frac{m}{s}+\frac{n}{t}=1$, m,n are constants and m < n. If the minimum value of (s + t) is $3+2\sqrt{2}$, then the equation of the chord of the ellipse $\frac{x^2}{4}+\frac{y^2}{16}=1$ whose mid-point is (m, n), is

(A) x + y = 3 (B) x – 2y + 3 = 0
(C) 2x + y = 4 (D) 4x + 2y = 3

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. The coefficient of the quadratic function f(x) including the constant term, are all rational has local maximum at x = 0. Let $g(x)=|f'(x)|e^{f(x)}$ has maximum value $4\sqrt{e}$. If $g(x)=4\sqrt{e}$ has rational solution then:

(A) $\int_{-1}^{0} g(x)dx = e-\frac{1}{e^7}$ (B) the value of $\text{sgn}(f(0))=-1$
(C) g(x) is non derivable at one value of x (D) the value of $g\left(\tan\frac{\pi}{4}\right)=\frac{2}{e^7}$

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42. A rod of length 2 units whose one end is (1, 0, –1) and other end touching the plane $x-2y+2z+4=0$, then

(A) the rod sweeps the figure whose volume is π cubic units

(B) area of the region which the rod traces out on the plane is 2π

(C) length of the projection of the rod on the plane is $\sqrt{3}$ units

(D) centre of the region which the rod traces out on the plane is $\left(\frac{2}{3},\frac{2}{3},-\frac{5}{3}\right)$

43. Let f(x) be a continuous function defined for every $x \in R$. For any real numbers a and b that satisfy $a<b, f(x)$ always satisfies $f(a)>f(b)$. Then which of the following is/are correct?

(A) $\lim\limits_{h\to 0}\frac{f(2+h)-f(2)}{h}$ exists and negative

(B) there is always only one real root of f(x) = 0

(C) there is always only one real root of f(x) = f(1 – x)

(D) there is no real root of f(x) = f(1 + x)

44. Let S be the set of all 3 × 3 matrices having 3 entries equal to 1 and 6 entries equal to 0. A matrix M is picked uniformly at random from the set S, then the correct statement(s) is(are)

(A) total number of matrices in the set S is 84

(B) probability that M is non-singular is $\frac{1}{14}$

(C) probability that M is identity matrix is $\frac{1}{14}$

(D) probability that M has trace equal to 0 is $\frac{5}{21}$

45. Let the function $f(x)=\frac{g^2(x)}{2}+3x^3-5$, where g(x) is a continuous and differentiable function. It is given that f(x) is a monotonically increasing function and g(0) = 4, then which of the following is true?

(A) $g^2(1)>10$ (B) $f(5)>3$

(C) $f\left(\frac{5}{2}\right)<2$ (D) $g^2(-1)<22$

46. For the equation $3^x=4^{x^2}$ the correct statements(s) is/are–

(A) number of solutions is 2

(B) number of solutions is 1

(C) one of the solution is $\log_4\left(\sqrt{3\sqrt{3\sqrt{3.....\infty}}}\right)$

(D) the sum of all the solutions is $\log_4\left(\left(1+\sqrt{2}\right)^2-2\sqrt{2}\right)$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. The solution set of the inequality $\left(\cot^{-1}x\right)\left(\tan^{-1}x\right)+\left(2-\frac{\pi}{2}\right)\cot^{-1}x-3\tan^{-1}x-3\left(2-\frac{\pi}{2}\right)>0$ is (a, b), then the value of $\left(\cot^{-1}a+\cot^{-1}b\right)$ is_____.

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48. Let λ and μ denotes the number of points of discontinuity and non-derivability of function $f(x)=\min\left(\{x^2\},\cos x\right)$ (where {.} denotes fractional part function), $x\in\left(0,\frac{\pi}{2}\right)$ then the value of $(\lambda+\mu)$ is equal to.....

49. Let f(x) be a continuous, periodic and bounded function with period 3 such that $\int_0^3 f(t)dt=6$. If $g'(x)=f(x)$, such that g(0) = 0, then value of $\lim\limits_{x\to 0} xg\left(\frac{1}{x}\right)$ is.....

50. Let a, b be the roots of the quadratic equation $x^2-(k-2)cx-(k-1)d=0$ and c, d be the roots of the quadratic equation $x^2-(k-2)ax-(k-1)b=0$. If a, b, c and d are distinct non-zero real numbers such that a + b + c + d = 100, then the integral value of k is_____.

51. Let $P\left(1,7,\sqrt{2}\right)$ be a point and line L is $2\sqrt{2}(x-1)=y-2,z=0$. If PQ is the distance of plane $\sqrt{2}x+y-z=1$ from point P measured along a line inclined at an angle of 45^0 with the line L and is minimum, then the value of PQ is.....

52. If cubic functions $f(x)=x^3+ax^2+bx+c$ and $g(x)=cx^3+bx^2+ax+1$ are such that $f(3)=0,g(4)=0$ and the $\lim\limits_{x\to p}\frac{f(x)}{g(x)}$ exists for all $p\in R-\{4\}$, then the value of $\lim\limits_{x\to -1}\frac{f(x)+g(x)}{x+1}$ is___.

53. Consider a curve passing through point (1, 1) such that perpendicular distance of normal drawn at any point P on the curve from origin is equal to ordinate of the point P. If area enclosed by the curve is A, then the value of [2A] is_____ (where [.] is greatest integer function)

54. An urn contains 3 red balls and n white balls. Mr. A draws two balls together from the urn and the probability that they have the same colour is $\frac{1}{2}$. Mr. B draws one ball from the urn, note down its colour and replaces it. He then draws a second ball from the urn and finds that the probability that both balls have the same colour is $\frac{1}{8}$, then possible value of n is_____.

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